# Lesson Objective

Interpret the unknown in multiplication and division to model and solve problems using units of 6 and 7.

## Materials

(S) Personal white board

| PROBLEM | TEACHER GUIDANCE |
|---|---|

1 **Model a multiplication word problem involving units of 6 by drawing and labeling a tape diagram, then solve by breaking the factor into two smaller facts.**

Rosa has 7 boxes of crayons. Each box has 6 crayons. How many crayons does Rosa have in all?

Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.

**Model a multiplication word problem involving units of 6 by drawing and labeling a tape diagram, then solve by breaking the factor into two smaller facts.**

Rosa has 7 boxes of crayons. Each box has 6 crayons. How many crayons does Rosa have in all?

Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.

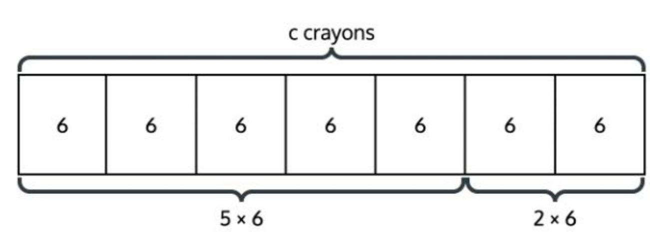
*Answer: c = 42 crayons; 7 × 6 = c; (5 × 6) + (2 × 6) = 30 + 12 = 42*

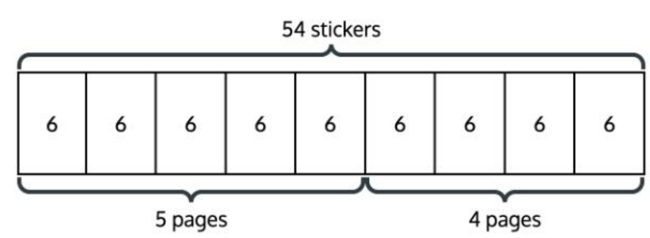
Tell students that we can draw a tape diagram to model this problem. Guide students through drawing a tape diagram with 7 equal parts, each labeled "6 crayons," and a bracket on top labeled "c crayons."

Ask students what the letter c represents in this problem. (The total number of crayons.)

Guide students to write the equation 7 × 6 = c. Ask students if they know a way to break 7 × 6 into two smaller facts they already know. Model partitioning the tape diagram into a group of 5 and a group of 2, and write (5 × 6) + (2 × 6).

Work with students to solve: 30 + 12 = 42. Have students write c = 42 crayons and a sentence to answer the problem.



| PROBLEM | TEACHER GUIDANCE |
|---|---|
| **2** **Model a division word problem involving units of 6 by drawing a tape diagram with an unknown number of groups, then solve using a related multiplication strategy or by decomposing the dividend.**<br><br>Marcus has 54 stickers. He puts 6 stickers on each page. How many pages does Marcus use?<br><br>Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.<br><br>*Answer: p = 9 pages; p × 6 = 54 or 54 ÷ 6 = p; (30 ÷ 6) + (24 ÷ 6) = 5 + 4 = 9* | Ask students what is different about this problem compared to the first one. (We know the total but not the number of groups.)<br><br>Guide students through drawing a tape diagram with the total of 54 stickers labeled on top, equal parts each labeled "6 stickers," and p representing the unknown number of pages.<br><br>Ask students to write an equation. (p × 6 = 54 or 54 ÷ 6 = p.)<br><br>Have students solve by decomposing the dividend: 54 ÷ 6 as (30 ÷ 6) + (24 ÷ 6) = 5 + 4 = 9. Ask students what p equals and have them write a sentence to answer the problem.<br><br><br> |
| **3** **Solve a word problem involving units of 7 by writing an equation with a letter for the unknown, drawing a model, and choosing a strategy to find the answer.**<br><br>A parking lot has 6 rows of cars. Each row has 7 cars. How many cars are in the parking lot?<br><br>Write an equation using a letter for the unknown. Draw a tape diagram and solve.<br><br>*Answer: t = 42 cars; 6 × 7 = t; (5 × 7) + (1 × 7) = 35 + 7 = 42* | Students work independently.<br><br>As students are ready, ask them to explain what their letter represents and how they broke apart the problem to solve. |

## Monitor For

- Does the student correctly identify what the unknown represents — the total, the number of groups, or the size of each group?
- Can the student draw and label a tape diagram that matches the structure of the word problem?
- Does the student decompose factors or dividends into smaller, known facts to solve?

## Instructional Moves

- If a student struggles to identify the unknown, ask: "What do you already know? What is the problem asking you to find?" and connect each quantity to the tape diagram.
- When decomposing, prompt students to use fives facts as a starting point: "What is $5 \times 6$? How many more groups do you need?"
- If a student's tape diagram doesn't match the problem structure, have them reread the problem and point to which part of the diagram shows each quantity.

## Reflection Questions

**1.** In Problem 1, the unknown was the total. In Problem 2, the unknown was the number of groups. How did knowing which quantity was unknown help you decide how to solve?
*Answer: When the total was unknown, I multiplied the number of groups by the size of each group. When the number of groups was unknown, I divided the total by the size of each group.*

**2.** How did you break apart the multiplication or division into smaller facts to solve?
*Answer: I used fives facts. For example, for $7 \times 6$, I did $5 \times 6 = 30$ and $2 \times 6 = 12$, then added $30 + 12 = 42$.*

**3.** How does the tape diagram help you see the structure of a word problem?
*Answer: The tape diagram shows the equal groups, the size of each group, and the total, so I can see which part is unknown.*

# Mini Lesson Student Materials

Name ___________________________________

**1.** Rosa has 7 boxes of crayons. Each box has 6 crayons. How many crayons does Rosa have in all?
Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.

**2.** Marcus has 54 stickers. He puts 6 stickers on each page. How many pages does Marcus use?
Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.

**3.** A parking lot has 6 rows of cars. Each row has 7 cars. How many cars are in the parking lot?
Write an equation using a letter for the unknown. Draw a tape diagram and solve.

# Mini Lesson Sample Script

## Materials

(S) Personal white board

| PROBLEM | SAMPLE SCRIPT |
|---|---|
| 1 **Model a multiplication word problem involving units of 6 by drawing and labeling a tape diagram, then solve by breaking the factor into two smaller facts.**<br><br>Rosa has 7 boxes of crayons. Each box has 6 crayons. How many crayons does Rosa have in all?<br><br>Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.<br><br>*Answer: c = 42 crayons; 7 × 6 = c; (5 × 6) + (2 × 6) = 30 + 12 = 42* | **T**: Read this problem. What do we know?<br><br>**S**: There are 7 boxes and each box has 6 crayons.<br><br>**T**: What are we trying to find?<br><br>**S**: How many crayons there are in all.<br><br>**T**: We can draw a tape diagram to model this. How many equal parts should we draw?<br><br>**S**: 7, because there are 7 boxes.<br><br>**T**: Right. Draw 7 equal parts on your board. What goes inside each part?<br><br>**S**: 6, because each box has 6 crayons.<br><br>**T**: Label each part "6." Now, the total is what we're trying to find. Let's use the letter c for the total number of crayons. Write "c crayons" above the whole tape.<br><br>**S**: (Label the tape diagram.)<br><br>**T**: Write an equation for this problem.<br><br>**S**: 7 × 6 = c.<br><br>**T**: Do you know 7 × 6 right away? Here's a way to break it into facts you might already know. What if we split the 7 groups into 5 groups and 2 groups? (Draw a line on the tape diagram between the 5th and 6th parts.) What is 5 × 6?<br><br>**S**: 30.<br><br>**T**: And 2 × 6?<br><br>**S**: 12.<br><br>**T**: So what is 30 + 12?<br><br>**S**: 42.<br><br>**T**: So c = 42 crayons. Write a sentence to answer the problem.<br><br>**S**: Rosa has 42 crayons in all. |

| PROBLEM | SAMPLE SCRIPT |
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2

**Model a division word problem involving units of 6 by drawing a tape diagram with an unknown number of groups, then solve using a related multiplication strategy or by decomposing the dividend.**

Marcus has 54 stickers. He puts 6 stickers on each page. How many pages does Marcus use?

Draw and label a tape diagram to model the problem. Write an equation using a letter for the unknown. Solve.

*Answer: $p = 9$ pages; $p \times 6 = 54$ or $54 \div 6 = p$; $(30 \div 6) + (24 \div 6) = 5 + 4 = 9$*

**S**: There are 54 stickers total, and each page gets 6 stickers.

**T**: What are we trying to find?

**S**: How many pages Marcus uses.

**T**: So this time, do we know the total or are we looking for it?

**S**: We know the total. We don't know how many groups.

**T**: Good. Let's use the letter p for the number of pages. On your tape diagram, label the top "54 stickers." Each part gets "6 stickers." We don't know yet how many parts there are — that's p. Write an equation.

**S**: $54 \div 6 = p$. → $p \times 6 = 54$.

**T**: Now, can we break 54 into parts that are easier to divide by 6? What's a number you know that divides evenly by 6?

**S**: 30. $30 \div 6 = 5$.

**T**: If we've used 30 of the 54 stickers, how many are left?

**S**: 24.

**T**: What's $24 \div 6$?

**S**: 4.

**T**: So how many pages is that altogether?

**S**: $5 + 4 = 9$ pages.

**T**: $p = 9$ pages. Write a sentence to answer the problem.

**S**: Marcus uses 9 pages.

| PROBLEM | SAMPLE SCRIPT |
| --- | --- |
| 3 **Solve a word problem involving units of 7 by writing an equation with a letter for the unknown, drawing a model, and choosing a strategy to find the answer.**<br><br>A parking lot has 6 rows of cars. Each row has 7 cars. How many cars are in the parking lot?<br><br>Write an equation using a letter for the unknown. Draw a tape diagram and solve.<br><br>*Answer: t = 42 cars; 6 × 7 = t; (5 × 7) + (1 × 7) = 35 + 7 = 42* | **T**: Now try this one on your own. Read the problem, write an equation with a letter for the unknown, draw a tape diagram, and solve.<br><br>**S**: (Work.) |